

Space Weather

RESEARCH ARTICLE

10.1029/2018SW002094

Key Points:

- GICs are estimated using the same network model at sites affected by the equatorial electrojet and the South Atlantic Magnetic Anomaly
- Both geomagnetic features increase GICs magnitude, with the greatest effects associated with the equatorial electrojet
- GICs with frequencies lower than 2 mHz are not amplified by the ionospheric currents

Supporting Information:

- Supporting Information S1
- Table S1

Correspondence to:

K. V. Espinosa,
karen.sarmiento@inpe.br

Citation:

Espinosa, K. V., Padilha, A. L., & Alves, L. R. (2019). Effects of ionospheric conductivity and ground conductance on geomagnetically induced currents during geomagnetic storms: Case studies at low-latitude and equatorial regions. *Space Weather*, 17, 252–268.
<https://doi.org/10.1029/2018SW002094>

Received 1 OCT 2018

Accepted 25 JAN 2019

Accepted article online 31 JAN 2019

Published online 8 FEB 2019

Effects of Ionospheric Conductivity and Ground Conductance on Geomagnetically Induced Currents During Geomagnetic Storms: Case Studies at Low-Latitude and Equatorial Regions

Karen V. Espinosa¹ , Antonio L. Padilha¹ , and Livia R. Alves¹ 

¹Instituto Nacional de Pesquisas Espaciais (INPE), São José dos Campos, Brazil

Abstract Geomagnetic field variations recorded by fluxgate magnetometers are used to evaluate geomagnetically induced currents (GICs) under the equatorial electrojet (EEJ) and South Atlantic Magnetic Anomaly (SAMA) during two geomagnetic storms in March and June 2015. Geomagnetic stations with information about the underground electrical conductivity structure and that can be approximated by unidimensional (1-D) models for calculation of the geoelectric field are selected. GICs levels are estimated using a realistic local power grid model located in the central region of Brazil, artificially moved to the sites where the geomagnetic measurements are available. GIC magnitudes are not large during the storms, and a maximum amplitude of 3.8 A was estimated at an equatorial station positioned over high resistivity underground during the main phase of the June storm. Effects of the ionospheric currents over the measurement sites and of the conductivity distribution beneath these sites are also evaluated. It is observed that both (EEJ) and (SAMA) increase the GICs amplitudes, with the greatest effects associated with daytime (EEJ) currents. A cutoff frequency was identified for both the (EEJ) and (SAMA) so that signals with frequencies lower than 2 mHz (periods longer than 500 s) are not amplified by the ionospheric currents. In relation to the underlying conductivity structure, GIC magnitude is affected by variation of the surface impedance and the sampling rate of the geomagnetic field. In our case, differences in amplitude of the surface impedance at the Nyquist frequency control the relative effects of the ground conductance between the stations.

1. Introduction

Geomagnetically induced currents (GICs) are driven in any grounded network of conductors due to rapid variations in the Earth's magnetic field related to extreme solar events. They may represent a significant risk in gas pipelines, telecommunication cables, railway signaling equipment, and, most importantly, power systems employed to generate and transmit electrical power (Boteler, 1994). Large GICs (around 100–500 A; Viljanen et al., 2013) are usually observed at high latitudes due to intense ionospheric current systems in the vicinity of the auroral ovals, and thus, their impact on grounded technological structures at these latitudes has been widely studied (Bolduc, 2002; Pirjola, 2000). In March 1989, a huge GIC reached the electrical transmission systems of Quebec, Canada, causing disruption of the power grid (Bolduc, 2002; Boteler, 1994). After that, many authors have measured/modeled GICs in different countries at middle and low latitudes, such as Brazil (Barbosa et al., 2015; Trivedi et al., 2007), South Africa (Bernhardi et al., 2008; Ngwira et al., 2008), China (C.-M. Liu et al., 2009), Japan (Watari et al., 2009), Spain (Torta et al., 2012), New Zealand (Marshall et al., 2012), Australia (Marshall et al., 2013), and Uruguay (Caraballo et al., 2013). These studies have shown that GICs may also be of concern to power networks at lower latitudes and some transformer failures have been reported (Gaunt & Coetzee, 2007). Other works have focused on the large rate of change of the geomagnetic field, especially in the equatorial region, which increase the susceptibility to GIC (Carter et al., 2015, 2016).

GIC magnitude in power networks depends not only on the engineering characteristics of the transmission line (length, linear resistance, earthing resistance, etc.) but also on the overhead ionospheric currents and on the underlying conductivity structure. Thus, GICs magnitudes can vary from tens to several hundred

of amperes depending on the latitude and strength of the magnetospheric activity. Higher GICs values are generally seen at higher latitudes, tending to peak for the largest storms around the auroral oval (Pulkkinen et al., 2012), while lower values are generally associated with regions of lower latitudes (e.g., Kappenman, 2003 and the references cited above for low-latitude studies). However, significant enhancements in the geomagnetic and geoelectric fields have also been reported at the equatorial region due to daytime equatorial electrojet (EEJ) currents (Carter et al., 2015).

South American longitudinal sector has the unique feature of the presence of the EEJ and the South Atlantic Magnetic Anomaly (SAMA), where energetic particle precipitation is observed in local ionosphere due to the global minimum intensity of the geomagnetic field. The well-known effect of the EEJ appears as a significant increase of amplitude in the horizontal component of a wide period range of geomagnetic fluctuations, which can be observed on the ground within a few degrees latitude around the dip equator (Forbes, 1981). At the SAMA, azimuthally drifting energetic particles trapped in the Earth's inner Van Allen radiation belt come closest to the Earth's surface, interacting with the dense atmosphere and producing enhanced ionization at ionospheric *E* layer heights (Paulikas, 1975). This enhancement in the ionization leads to the rise of the ionospheric conductivity, which is a regular feature of the ionosphere over the SAMA even under magnetically quiet conditions that can be further enhanced during magnetically disturbed periods (Abdu et al., 2005). Enlarged amplitudes observed in magnetic horizontal component at the center of the SAMA during storm sudden commencement (SSC) have been ascribed to the increase of electron precipitation in this region (Trivedi et al., 2005). Considering the contribution of ionospheric currents to the GIC magnitude in high latitudes, it is expected that the low-latitude ionospheric effects associated with the EEJ and the SAMA can also contribute to enhance the GIC intensity at these locations.

Besides the local features of ionospheric currents, calculation of the spatial distribution of the geoelectric field on the Earth's surface requires knowledge of the conductivity structure down to depths of several tens or a couple of hundred kilometers underneath the study area. Because the intensity of the induced electric field is inversely proportional to conductivity, higher GICs intensity are expected to be generated in low conductivity (i.e., resistive) regions. Information about the underground conductivity can be obtained by employing deep-probing electromagnetic techniques, such as magnetotellurics (MT). The simplest way to determine the geoelectric field is to assume that geomagnetic variation is a plane wave and that the conductivity of the Earth only varies with depth in a unidimensional (1-D) model. This is the basic model in MT and results in spatially constant horizontal electric and magnetic fields on the Earth's surface (Pirjola, 2002). The plane wave assumption could be questionable under nonuniform EEJ and SAMA ionospheric currents but MT experiments around the dip equator in Brazil have shown that this assumption is valid at least up to periods of 34 min (Padilha et al., 1997). It must be observed that periods longer than about 30 min that could be affected by the source field geometry different from plane wave do not produce large GICs (Trivedi et al., 2007). On the other hand, it is known that the electrical conductivity of the crust and mantle can vary strongly in all three spatial dimensions and that using a single 1-D model neglects variations of the geoelectric field due to these lateral changes in conductivity. Several recent publications have illustrated the limitations of using the 1-D approximation for geoelectric field estimation when compared to realistic three-dimensional (3-D) Earth impedances (e.g., Weigel, 2017). However, applying a 3-D approach requires a dense array of MT measurements, currently only available in parts of the United States and Australia. To circumvent this problem, it is possible to choose sites where simple layered-Earth models are reasonable approaches to represent the local conductivity distribution (Pirjola, 2002).

In this paper, GICs are estimated for different sites in Brazil during the two largest geomagnetic storms occurred in 2015. We use 1-s time series recordings from geomagnetic stations of the Embrace Magnetometer Network (Embrace MagNet; Denardini et al., 2018) converted to 1-min averages. Among the available geomagnetic stations, we selected those with information on underground electrical conductivity structure and that can be approximated by 1-D models for calculation of the geoelectric field. GICs levels are estimated using a realistic local power grid model situated in the central region of Brazil (Barbosa et al., 2015), artificially moved to the sites where the geomagnetic measurements are available, with the currents always calculated in the same substation of the network. The results are here interpreted in terms of the ionospheric currents above the measurement sites and the conductance distribution (related to the product between the conductivity and the layer thickness for each 1-D layer) beneath these sites.

2. Methodology

The physical concept of the GIC flow is associated with the Faraday's induction law, which relates the temporal variation of the geomagnetic field to the induction of the geoelectric field. The geoelectric field induced within the Earth then triggers GIC in ground systems according to Ohm's law. Estimation of GICs in a particular network node can be divided into two independent steps (Pirjola, 2002): (1) determination of the horizontal geoelectric field at the Earth's surface and (2) computation of GICs using this geoelectric field and engineering parameters of the power network.

A practical way to calculate the geoelectric field is to use geomagnetic data and a relation between geoelectric and geomagnetic fields. This requires spatial and temporal measurements of the geomagnetic field and a model of the conductivity structure of the region. The simplest geoelectric field model assumes that a plane wave is propagated vertically downward and that the Earth is composed of a horizontally layered structure with various thickness h and conductivity σ (Cagniard, 1953; Tikhonov, 1950). The geoelectric field components $E_{x,y}$ can be computed in the frequency domain in terms of the perpendicular geomagnetic field components $B_{y,x}$ as

$$E_{x,y}(\omega) = \pm \frac{Z(\omega)}{\mu_o} B_{y,x}(\omega), \quad (1)$$

where $Z(\omega)$ is the surface impedance dependent of the ground conductivity, ω is the angular frequency, μ_o is the vacuum permeability, and $B_{y,x}(\omega)$ are the Fourier transformed magnetic field components in the frequency domain.

The complex surface impedance $Z(\omega)$ can be calculated using the Wait's recursion relation (Simpson & Bahr, 2005; Wait, 1954) that allows the impedance at one interface (Z_n) to be derived from the impedance at the interface directly below (Z_{n+1}).

$$Z_n(\omega) = \frac{\omega\mu_o}{k_n} \left[\frac{k_n Z_{n+1} + \omega\mu_o \tanh(k_n h_n)}{\omega\mu_o + k_n Z_{n+1} \tanh(k_n h_n)} \right], \quad (2)$$

where h_n is the thickness of the n th layer, k_n is given by $(1-i)\sqrt{\omega\mu_o\sigma_n/2}$ for the n th layer with conductivity σ_n and for the deepest semi-infinite half-space layer $Z_n(\omega) = \sqrt{i\omega\mu_o/\sigma_n}$.

The geoelectric fields $E_x(t)$ and $E_y(t)$ in the time domain are then computed from the inverse Fourier transforms of $E_x(\omega)$ and $E_y(\omega)$, respectively.

In the second step the currents flowing in the grounded technological system are estimated based on the determined geoelectric field and detailed information about the topology and electrical properties of the particular ground system. Following Kirchhoff's laws, the power grid can be modeled as a set of n nodes (network substations). Owing to the low GIC frequency compared to 50/60 Hz, a direct current treatment is sufficient and Lehtinen and Pirjola (1985) derive the following formula for the earthing currents at each node:

$$I = (U + YZ)^{-1}J, \quad (3)$$

where I is a vector containing the earthing currents at each node, J is a perfect earthing current, and U is the $n \times n$ identity matrix. The matrices Z and Y represent the earthing impedance and network admittance matrix, respectively. The elements of the admittance matrix are defined by the resistance of the conductors in the network:

$$Y_{ij} = \frac{-1}{R_{ij}} \quad i \neq j \quad Y_{ij} = \sum_{\substack{k=1 \\ k \neq i}}^n \frac{1}{R_{ik}} \quad i = j, \quad (4)$$

where i and j represent the nodal points of the earthing transformers that integrate the substations and interconnect the electric power transmission network.

2.1. Geomagnetic Data

Geomagnetic field variations in South America are available from the Embrace MagNet. Data are acquired by commercial fluxgate magnetometers with 1-s sampling rate and details of sensitivity and calibration procedures can be found in Denardini et al. (2018). One-minute values were calculated for this study as averages of 60 one-second values, centered on the minute.

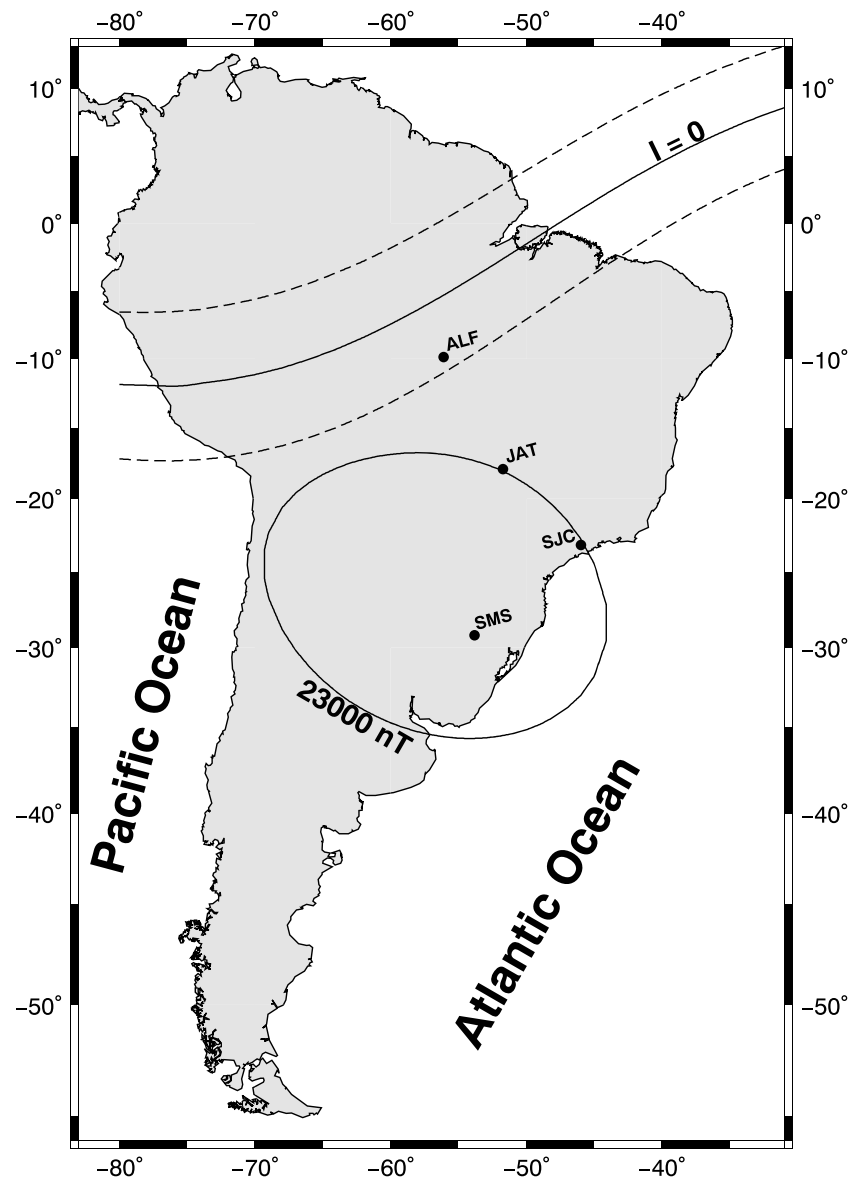


Figure 1. Location of the four Embrace MagNet stations used in this study and the areas of influence of the equatorial electrojet and South Atlantic Magnetic Anomaly during the year 2015. The solid line shows the dip equator ($I = 0$) and the dashed lines represent $\pm 10^\circ$ dip angles. The 23,000-nT isoline of the total intensity of the geomagnetic field is used to outline the South Atlantic Magnetic Anomaly.

We calculate GIC during two intense geomagnetic storms ($Dst < -200$ nT) occurred in the decreasing phase of the sunspot cycle 24, on 17 March and 22 June 2015. These are the largest recorded geomagnetic storms since the Embrace Mag-Net became operational. Among the 12 geomagnetic stations currently in operation, 4 were chosen based on data availability for these events and knowledge of the ground conductance structure. Figure 1 shows a map with the geographic location of these stations along with contours of relevant geomagnetic coordinates during the period of data analysis. These include the position of the dip equator, inclination of $\pm 10^\circ$ that delimit the region where the EEJ currents are expected to affect geomagnetic variations recorded on the ground (Padilha et al., 2003) and an isoline of 23,000 nT of the total magnetic field intensity that encompasses the position of the SAMA center. Details on the stations are presented in Table 1, where the geomagnetic information was obtained using the AACGM-v2 software (Shepherd, 2014) and IGRF-12 coefficients (Thébault et al., 2015).

Table 1*Geographic and Geomagnetic Coordinates and Total Geomagnetic Field Strength at the Selected Embrace MagNet Stations for the Year 2015.*

Station code	Geographic		Geomagnetic		Total field (nT)
	Latitude (°)	Longitude (°)	Latitude (°)	Longitude (°)	
ALF	−9.87	−56.10	−2.82	16.18	24,251
JAT	−17.93	−51.72	−12.61	17.83	23,022
SJC	−23.21	−45.96	−19.53	21.47	22,982
SMS	−29.44	−53.82	−21.61	13.56	22,474

ALF is an equatorial station located in the region under influence of the EEJ diurnal currents. SMS has the lowest intensity of the geomagnetic field and therefore is the station closest to the center of SAMA. Following previous results (Trivedi et al., 2005), it is expected to be the station most affected by the increased ionospheric conductivity due to particle precipitation in the anomaly. The stations JAT and SJC have almost the same total magnetic field intensity (difference of 40 nT) and thus are relatively equidistant from the center of the anomaly. However, this does not guarantee that ionospheric effects are similar in the two stations, because particle precipitation is not symmetric around the SAMA center (Abdu et al., 2005).

To address this latter issue, the SSUSI sensor onboard the Defense Meteorological Satellite Program (DMSP) satellites is useful for tracking particle precipitation at the top of SAMA (Paxton et al., 2002). The sensor observes the atmospheric emission at 427 nm, which is proportional to the combined effect of proton and electron precipitation flux at the spacecraft's altitude (~840 km). Figure 2 shows the relative intensity of the particle precipitation fluxes throughout the SAMA region during the two geomagnetic storms as measured by the SSUSI sensor. At DMSP altitude, the local maximum of the particle flux is positioned just off the Brazilian coast, close to the location of the SJC station on the ground. It can be observed that this area of maximum particle flux has different size and location in different energy bands. Grigoryan et al. (2005) have shown that when the particle energy increases, the maximum of the particle flux approaches the minimum of the geomagnetic field. For high-energy particles capable of penetrating to E layer heights (90–120 km) it is expected that the maximum flux is much closer to the minimum of the geomagnetic field intensity, used here to define the SAMA center. In addition, the oval shape of the region of maximum particle precipitation differs significantly from that used to represent the SAMA based on the total intensity of the geomagnetic field (Figure 1).

2.2. Conductivity Models

Simplified 1-D conductivity models for the regions where the geomagnetic stations are located are based on interpretations of past MT measurements. These models are shown in Table 2, where the Earth is modeled with multiple horizontal layers of different resistivity and thickness so that the resistivity changes vertically but not horizontally.

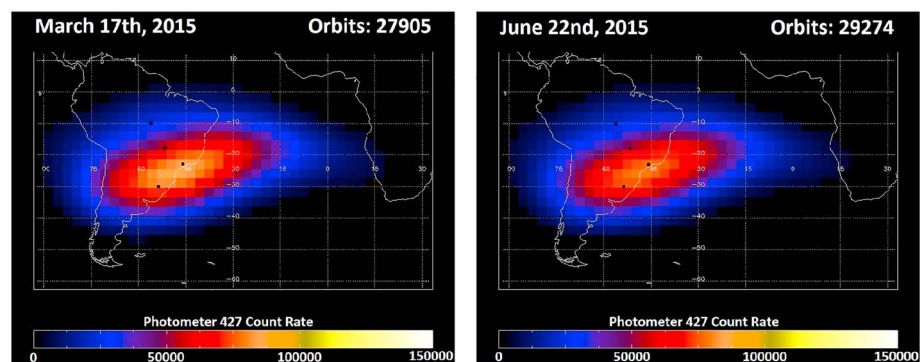


Figure 2. Atmospheric fluorescence at 427 nm registered by the SSUSI sensor onboard the Defense Meteorological Satellite Program satellites over the South Atlantic Magnetic Anomaly region during the 2015 storms. The photon count rate is proportional to the intensity of the charged particle radiation flux. Black dots show the position of the ground geomagnetic stations.

Table 2
The 1-D Resistivity Models Below Each Geomagnetic Station From Magnetotelluric Surveys

Station		Number of layers						
		1	2	3	4	5	6	7
ALF	Resistivity (Ωm)	200	10,000	1,100	7,000	300		
	Thickness (km)	1	10	20	230	—		
JAT	Resistivity (Ωm)	60	20	80	7,000	50	5,000	300
	Thickness (km)	0.5	0.7	0.6	42	36	120	—
SJC	Resistivity (Ωm)	5	10,000	100	10,000	5,000	300	
	Thickness (km)	0.2	10	2	10	200	—	
SMS	Resistivity (Ωm)	160	12	5,000	500	5,000	300	
	Thickness (km)	0.2	1	20	20	160	—	

ALF is located in the southwestern region of the large Amazonian craton, consisting of an ancient Archean core and adjoining Paleoproterozoic and Mesoproterozoic provinces. MT soundings performed in this region (Fernandes et al., 2018) indicate a highly resistive structure down to upper mantle depths (considering here an average crustal thickness of 40 km). JAT and SMS are respectively positioned on the northeastern and southern edges of the intracratonic Paraná basin, where Paleozoic sediments are covered by an extensive Early Cretaceous basalt complex. In the region of the JAT station, the conductivity model is derived from Bologna et al. (2011) and shows a resistive crust beneath the basin, underlined by a conductive upper mantle. In SMS, where the geoelectric model is obtained from Dragone (2018), the crust is much more conductive and the upper mantle is much more resistive than in JAT. SJC is located on the Tertiary Taubaté basin, whose sedimentary package consists mainly of shales and sandstones, with intercalated clay units. The basin was extensively studied by MT and audiomagnetotelluric soundings and a simplified 1-D geoelectric section presents a highly conductive sedimentary package underlying a resistive lithosphere (thickness of approximately 200 km) but with a conductive layer in the middle crust (Padilha et al., 1991). On the other hand, this geomagnetic station is positioned at a distance of about 70 km from the Atlantic Ocean coastline, and therefore, the coast effect is a major concern in proposing a local 1-D model. However, detailed modeling considering the subsurface geology, contrast in conductivity between seawater and subsurface layers, depth of the ocean, and distance from the basin to the coast showed that such an effect is insignificant within the basin for the typical GIC periods (Brito, 1998).

The surface impedance values for the regions of the geomagnetic stations can be calculated using the 1-D Earth resistivity models as inputs to the recursive relation of equation (2). Figure 3 shows the amplitudes and phases of the surface impedances as a function of the frequency ($f = \omega/2\pi$). As we are mainly interested in the magnitudes of the geoelectric field, the absolute values of the surface impedances ($|Z(\omega)|$) are more important. It can be seen in all models that the amplitudes increase with frequency, indicating that the Earth behaves like a high-pass filter. As the larger GICs amplitudes are associated with rapid variation rates of the geomagnetic field, the main influence of the underground conductivity is concentrated in upper crustal layers due to the lower depth of penetration of the high-frequency signals. This high-frequency part

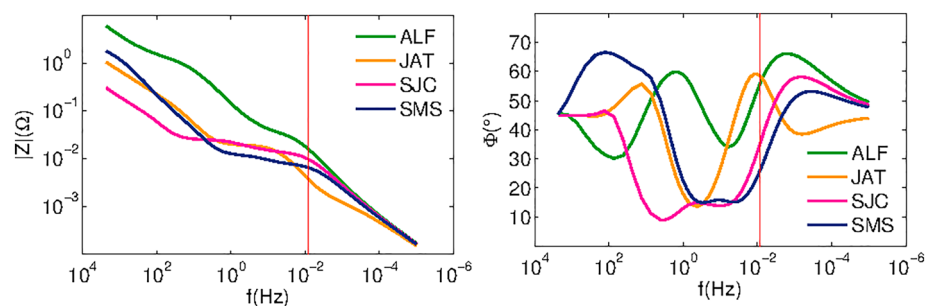


Figure 3. Amplitude (left panel) and phase (right panel) of the surface impedances for the locations of the four geomagnetic stations based on the 1-D conductivity models. Vertical red lines represent the Nyquist frequency (~ 8.3 mHz) considering the sampling frequency of the geomagnetic data.

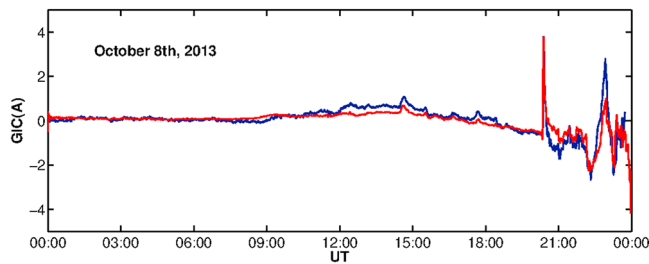


Figure 4. Comparison between calculated (red) and measured (blue) geomagnetically induced currents (GICs) in the neutral of a power transformer of a substation in central Brazil during a geomagnetic storm on 8 October 2013. Measured values are available in the supporting information.

is affected by layers closer to the surface, which present a wide variety of regional conductivities and cause the large differences observed between the models in Figure 3. On the other hand, the amplitudes are almost the same in the low-frequency part because at these frequencies the electromagnetic waves penetrate deeper into the Earth and are affected mainly by the deeper layer with the same conductive value.

The figure also shows the Nyquist frequency considering the acquisition rate used for the geomagnetic data (1 sample per minute). For the frequency interval of interest, the surface impedance is already affected by the high conductivity of the mantle. The highest absolute values of the impedance at these frequencies are observed in ALF, and thus, the effect of ground conductance is to increase GIC amplitude at this station in comparison with the others. SJC and SMS have almost the same impedance amplitude, and therefore, the Earth conductivity effect should

be similar. The impedance amplitude of these three stations is very close for frequencies approaching 1 mHz. JAT is much more conductive at upper mantle depths, which results in smaller impedances and hence a damping of low-frequency GICs is expected at this station.

2.3. Power Network

Viljanen et al. (2013) considered the model of a fixed standard grid (a Finnish 400-kV power), intensity of geomagnetic variation, and conductivity profiles to characterize GIC in different locations in Europe. A similar procedure is developed here by using a typical Brazilian power grid to provide a reference for estimating GICs at the sites where geomagnetic measurements and ground conductance distribution are available.

The power grid model comes from a 500-kV network operated in central Brazil and composed of seven earthed substations. In the absence of public information on the network parameters, we assumed some of the values suggested in Barbosa et al. (2015) and derived others that provided the best fit to measured GICs. The line resistance per unit length was assumed to be $0.02 \Omega/\text{km}$, and a common value of 0.43Ω was chosen for all earthing impedances. These earthing impedances are the sum of the transformer resistances and the actual earthing resistances. We also represent all nodes as simple conductors, which means that a three-phase power transmission line is connected to a transformer with a neutral point into the ground. For calculation purposes, this power grid was considered to operate in different sites in Brazilian territory by modifying its coordinates but keeping unchanged the relative locations of the nodes and the lengths of the transmission lines.

Direct measurements of GICs in the neutral phase of the power transformer at a substation of this grid are available for the period 2009–2014. These results can be used to validate our procedure for estimating GIC through geomagnetic data. We follow the same procedure described in Barbosa et al. (2015) that calculated the GIC amplitudes in the grid during a geomagnetic storm occurred on 8 October 2013. Geomagnetic variations are available from the Vassouras magnetic observatory, and ground conductance distribution was given by Trivedi et al. (1997). Figure 4 shows a comparison between measured and calculated GIC in the power grid transformer during the geomagnetic storm. It can be seen that GIC peaks are in good agreement (maximum amplitude of 4 A), an indication that the calculation procedure is reliable.

3. Geomagnetic Storms of 2015

The two largest geomagnetic storms of solar cycle 24 occurred in 2015, on 17 March and 22 June, with minima Dst of -223 and -204 nT, respectively. Both storms are characterized by multistep development, which is usually accompanied by a large rate of change in the geomagnetic field that acts as sources that could drive enhanced GICs. Sudden events, such as sudden impulse or SSC related to interplanetary shocks, are important disturbance sources of GIC (Kappenman, 2003), and there are evidences at low-latitude locations that GIC due to SSC are much higher than during the main phase (Zhang et al., 2015).

The St. Patrick's Day storm of 17 March 2015 (Carter et al., 2015) had two interacting coronal mass ejections (CMEs) associated with two flares from the same Sun's active region (Y. D. Liu et al., 2015). These CMEs hit the Earth's magnetosphere and generated a SSC recorded by ground observatories at around 04:44 UT on 17

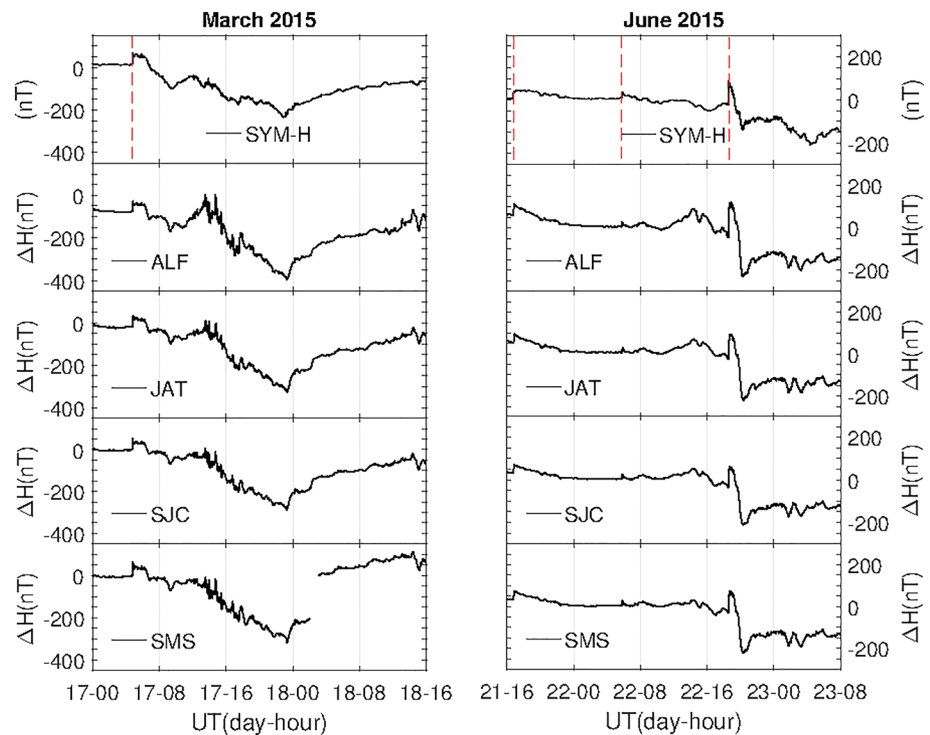


Figure 5. Time series of the magnetic field H component at the four Embrace MagNet stations and the $SYM-H$ index during the geomagnetic storms of March 2015 (left panels) and June 2015 (right panels). Vertical red-dashed lines on the $SYM-H$ index graphs indicate the times of the storm sudden commencement for the March storm (04:44 UT on 17 March) and the three shocks for the June storm (16:43 UT on 21 June, 05:44 UT and 18:32 UT on 22 June).

March 2015. It was characterized as a typical two-step storm, driven by two successive southward interplanetary magnetic field (IMF) structures, which led to an intense particle precipitation and an enhancement in substorm activity from 06:00 to 09:00 UT and from 12:00 to 23:00 UT on 17 March. The storm is well studied by many researchers and details on extreme local variations of the geomagnetic field associated with magnetosphere-ionosphere processes during this event can be found in several papers (e.g., Astafyeva et al., 2015; Kamide & Kusano, 2015; Ngwira et al., 2018).

The storm of 22 June 2015 was associated with an active region on the Sun that produced a sequence of CMEs associated to strong flares, from 18 to 25 June (Y. D. Liu et al., 2015). The geomagnetic disturbances at Earth were caused by three interplanetary shocks that impacted the magnetosphere in 21 June at around 16:43 UT and 22 June at around 05:44 and 18:32 UT. The first shock did not increase the auroral activity, while the second caused an enhancement in the substorm activity. The third shock was accompanied by a sudden increase in the solar wind speed due to amplification by the preceding shocks and a large southward IMF (B_z) component that triggered a great geomagnetic storm. The event was not as well studied as that of March, but more details can be found in Astafyeva et al. (2016).

The magnetic signature of the two geomagnetic storms is shown in Figure 5, which plots the H component of the geomagnetic field as recorded at the four Embrace MagNet stations considered. As the ground geomagnetic variations are strongly polarized to the H component at low and equatorial latitudes, these are the disturbances associated to the generation of GIC. The H -symmetric ($SYM-H$) index, representing the variation of the H component using globally distributed low-latitude magnetic observatories, is also included for comparison. In these graphs, the times of the SSC for the March storm and of the three shocks for the June storm are identified by red-dashed lines. There is a gap of 1 hr in the geomagnetic data recorded in the SMS station during the recovery phase of the March storm. However, this does not complicate the GICs calculation because its amplitudes are much larger during the SSC and main phase of the storm.

On 17 March, the SSC was observed at nighttime in the South American sector (01:44 LT) and its maximum amplitude was recorded at the station closest to the SAMA center (SMS, amplitude of ~ 63 nT). During the main phase of this storm, the minimum value was recorded at the ALF equatorial station (~ -396 nT at

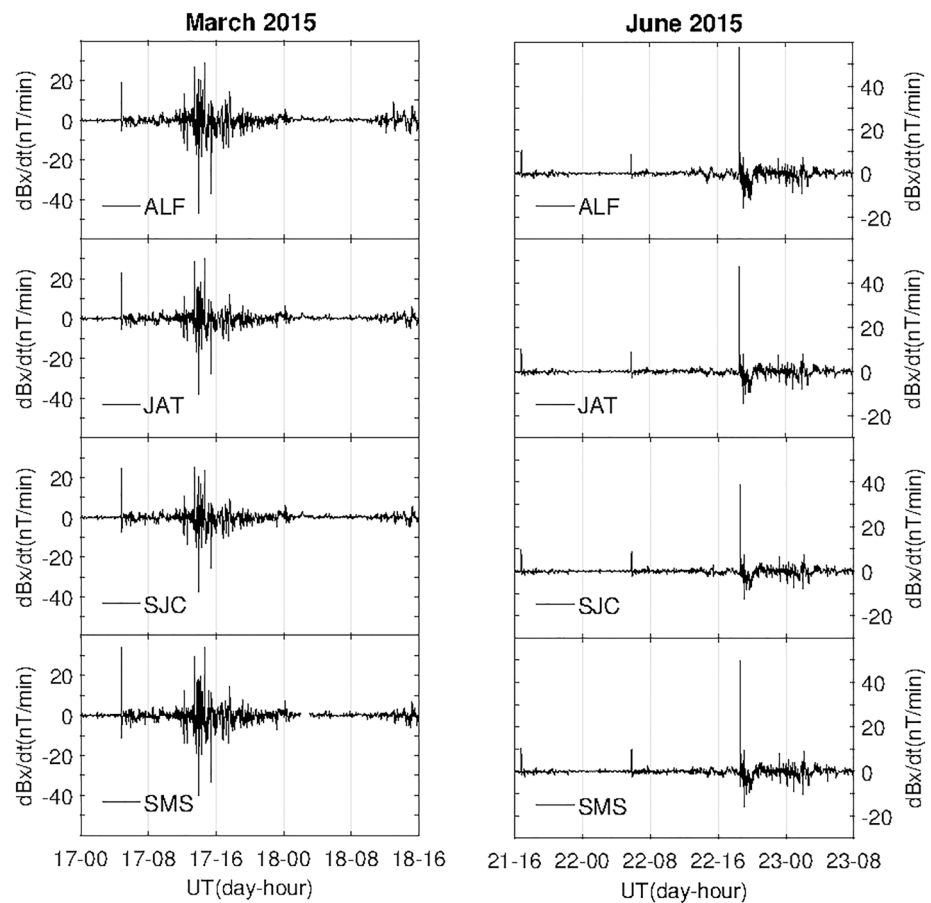


Figure 6. The time derivatives of the X component of the ground geomagnetic field at the Embrace MagNet stations during the March and June geomagnetic storms.

20:13 LT on 17 March). For the June storm, the first shock was recorded during daytime (13:43 LT), with maximum amplitude at the ALF equatorial station (~ 53 nT). The two subsequent shocks on 22 June took place during nighttime (02:44 LT) and daytime (15:32 LT). The minimum value during the main phase was also recorded at the ALF station (~ -231 nT at 17:15 LT on 22 June).

In general, the largest GICs are expected when large rates of change of the geomagnetic field (dB/dt) are observed. Before obtaining the time derivatives, we converted the horizontal (H) and declination (D) geomagnetic data of the Embrace MagNet stations into northward (x) and eastward (y) data, following $B_x = H \cos D$ and $B_y = H \sin D$. As a typical result for low magnetic latitudes, the time derivative of the X component is much larger than the Y component. Figure 6 provides a comparison of the dB_x/dt obtained at the four geomagnetic stations during the two geomagnetic storms.

The maximum dB_x/dt during the nighttime SSC event of the March storm was observed at the SMS station (~ 34 nT/min). However, the maximum dB_x/dt value for this storm occurs during the daytime main phase (10:54 LT) at the ALF station (~ -47 nT/min). From previous studies (e.g., Pulkkinen et al., 2003), this sharp increase in the rate of change of the geomagnetic field during the main phase of the storm can be associated with the onset of substorm activity, which produces large temporal variations in auroral ionospheric current systems. The penetration of high-latitude convection electric fields of magnetospheric origin (called prompt penetration electric fields) can be invoked to generate the geomagnetic disturbances in equatorial and low-latitude regions during the geomagnetic storm (Kikuchi et al., 1978).

The first two shocks of the June storm produced small spikes in the geomagnetic field (time derivatives around 10 nT/min). On the other hand, the third shock generated the highest dB_x/dt recorded by our geomagnetic stations for the year 2015. This shock occurred during daytime in the South American sector (15:32 LT), and the maximum rate of change was observed in ALF (~ 58 nT/min). Despite not having the

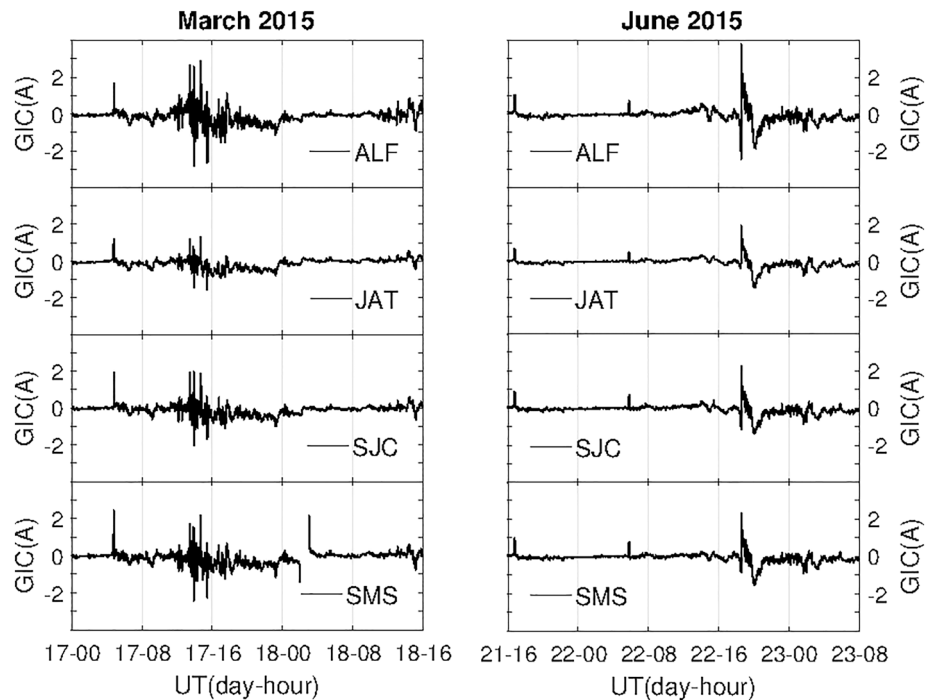


Figure 7. Time series of the estimated geomagnetically induced current (GIC) flow at the geomagnetic stations during the two storms.

largest Dst , the June storm produced the highest dB/dt intensity in our stations associated with the arrival of an interplanetary pressure pulse just behind two other pulses and a strongly negative IMF (B_z). This result is expected for low-latitude regions, where the maximum dB/dt values often coincide with the abrupt onset of the storm rather than during its main phase (Kappenman, 2003).

4. GICs Estimates

The estimated GIC flow at the four stations during the two geomagnetic storms is shown in Figure 7. Following the procedures described in section 2, the observed dB_x/dt and the ground conductance model for each station were first used to determine the ground electric field. Using the reference power grid parameters, the GIC was then modeled based on this electric field estimate.

The results show that the largest GICs amplitudes are generally observed in the ALF equatorial station. The only exception is for the SSC of the March storm, which occurs at nighttime and therefore has the highest dB_x/dt in SMS. In this case, the maximum amplitude at this station is of ~ 2.4 A. However, the largest amplitude for this storm is observed in SMS during the daytime main phase (14:41 UT on 17 March), with an estimated value of ~ 3 A. For the June storm, the first two shocks have very low amplitudes (maximum around ~ 1 A in ALF), but the third shock generates the largest GIC modeled in this study. The GIC reaches ~ 3.8 A in ALF at 18:32 UT on 22 June, while at the other stations the value is around 2 A. In addition to the high subsurface resistivity in ALF, the higher dB_x/dt associated with the daytime EEJ amplification also contributes to this result.

5. Effects of Overhead Ionospheric Currents and Underlying Conductivity Structure

For a more detailed analysis of the differences between the estimated GIC amplitudes at each station, Figure 8 shows its temporal evolution during the main phase of both storms. To quantify the differences, a minimum threshold value was chosen so that only individual peaks with GICs amplitudes outside the range of ± 0.4 A are considered. These peaks are used to calculate relative values between SMS (taken here as reference) and the other stations. They are shown in the lower panels, with relative values above 1 (identified with a red line) indicating that the event at SMS is amplified in relation to the other station.

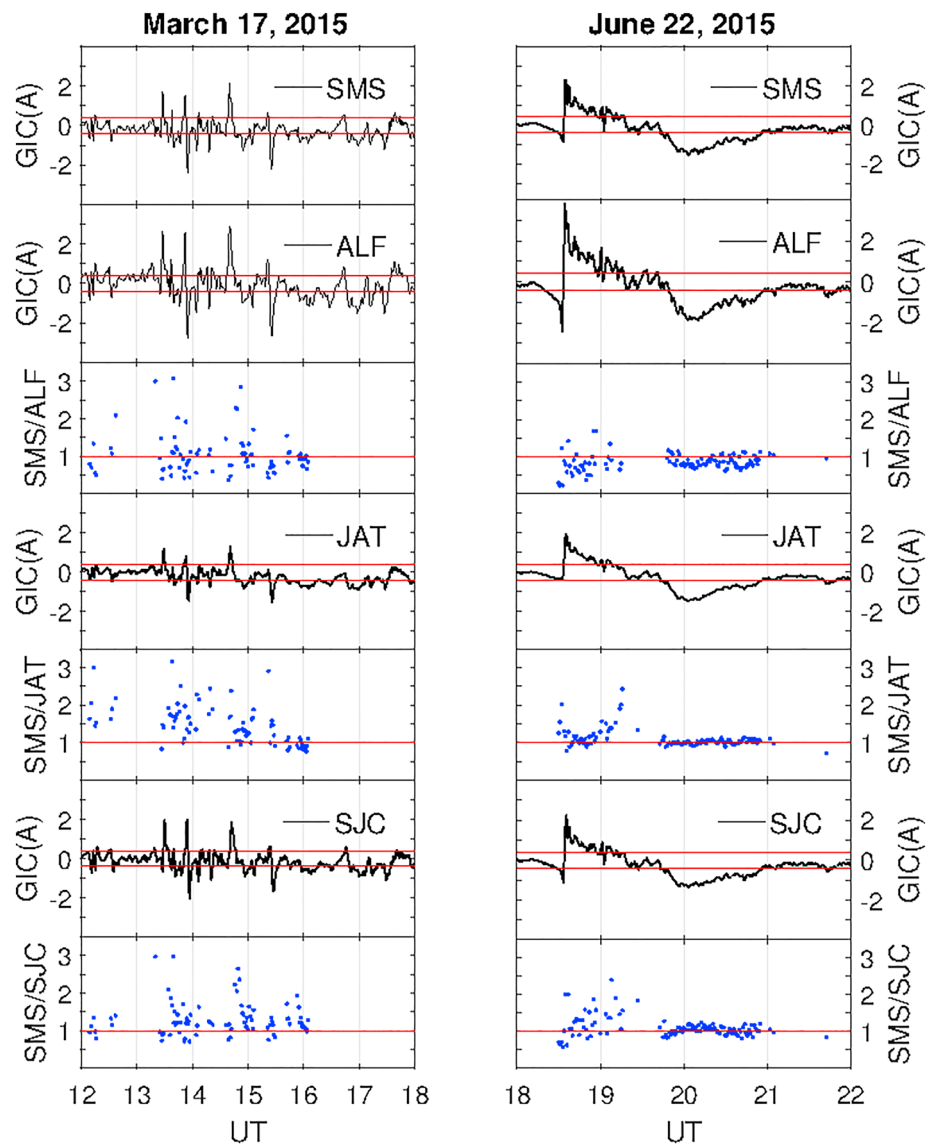


Figure 8. Absolute and relative geomagnetically induced currents (GICs) amplitudes during the main phase of the March and June storms. The top panels show the time evolution of the estimated GIC in the stations SMS, ALF, JAT, and SJC. Red lines at ± 0.4 A are used as the minimum threshold for GICs peaks. The bottom panels show relative values (blue dots) between SMS and the other stations for the GICs peaks.

In general, the peaks with higher amplitude are larger in ALF than in SMS, which is most clearly seen in the higher GICs of the June storm (time interval roughly between 18.5 and 19.5 UT). For these same events, the amplitudes in JAT and SJC are markedly smaller than in SMS. An interesting behavior is observed in the low-frequency GICs estimated during the June storm (approximately between 20 and 21 UT), whose amplitudes are approximately the same in all stations. All these results are influenced simultaneously by ground conductance and overlying ionospheric currents at each station, but it is not possible to separate the influence of each effect in Figure 8.

An approximate estimate of how much magnetic field variations alone can affect GIC amplitudes is to calculate GIC at each site using the same ground conductance model (i.e., only the magnetic field changes at each site). This procedure is often done in studies that use Quebec's highly resistive 1-D model and British Columbia's more conductive 1-D model derived by Boteler and Pirjola (1998) to assess how geomagnetic variations alone drive GICs in different locations. For this exercise, the JAT conductivity model was chosen to be the reference in the calculation of the horizontal geoelectric fields. Figure 9 shows the estimated GIC in each of the stations using the impedance for JAT (identified as Z_{JAT}) and the geomagnetic variation in

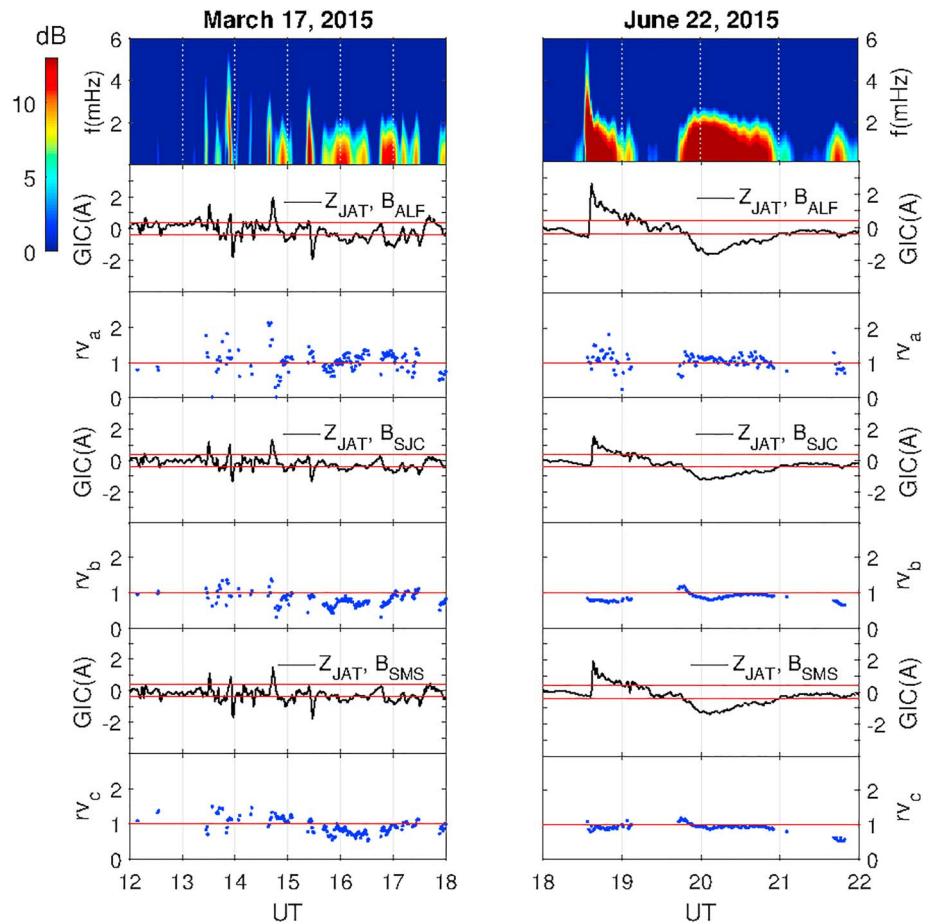


Figure 9. Absolute and relative geomagnetically induced currents (GICs) amplitudes during the main phase of the March and June storms using the same impedance as reference (Z_{JAT}). Time evolution of the estimated GIC using the magnetic field in stations ALF, SJC, and SMS are shown. Relative values (blue dots) are calculated between these estimated GICs and those of JAT in Figure 8. The top panels show spectrograms of the GICs in station JAT.

the three other stations (B_{ALF} , B_{SJC} and B_{SMS}) during the main phase of both storms. To quantify the effects of the geomagnetic variation, GIC peaks with amplitudes above the reference value (± 0.4 A) were used again to calculate relative values between the three stations and JAT (in this case using the results shown in Figure 8). To determine exactly which frequencies are contributing to the magnetic signal, spectrograms of the GICs in JAT are also included.

The spectrogram for the March storm shows the occurrence of several high-frequency GICs peaks (frequencies up to 5 mHz) during the first phase of the storm (approximately between 13.3 and 15.5 UT). The second phase of the storm (after 15.5 UT) is characterized by low-frequency GICs (lower than 2 mHz). In the June storm, there is an initial high-frequency peak related to the third shock around 18.5 UT, but the rest of the storm, especially after 19.7 UT, is marked by GICs with frequencies lower than 2 mHz.

The relative values considering the difference between the magnetic fields in ALF and JAT show an amplification in ALF during the intervals characterized by high-frequency signals in both storms. This indicates that the EEJ acts in the amplification of GICs in these high-frequency bands. On the other hand, for the low-frequency intervals of both storms, GIC values are approximately the same. This indicates that there is no effect of the EEJ in these frequencies. In the comparison between the effects of the magnetic fields in SJC and JAT, the relative values are approximately identical (ratios oscillate around 1) in the high-frequency interval of the March storm. JAT generally has higher values for the other phases of the two storms. In the comparison between SMS and JAT, the high-frequency phase of the March storm presents higher values in SMS but with amplitudes lower than those observed in the comparison with ALF. This indicates that the high-frequency signals are amplified in the vicinity of the SAMA center, although with less amplification

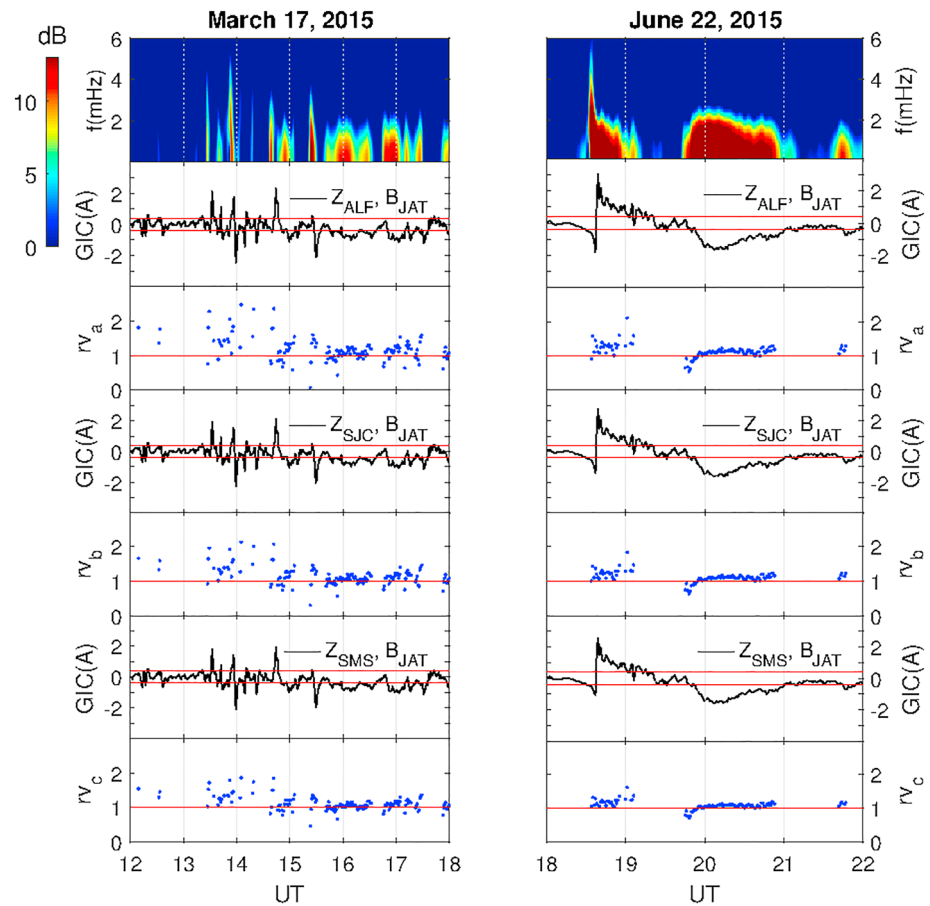


Figure 10. Absolute and relative geomagnetically induced currents (GICs) amplitudes during the main phase of the March and June storms using the same magnetic field as reference (B_{JAT}). Time evolution of the estimated GIC using the impedance of stations ALF, SJC, and SMS are shown. Relative values (blue dots) are calculated between these estimated GICs and those of JAT in Figure 8. The top panels show spectrograms of the GICs in station JAT.

than in the EEJ region. In the low-frequency intervals, GICs have similar amplitude, indicating that there is no effect of SAMA at these frequencies.

A similar procedure was adopted to estimate how much changes in ground conductance are relatively effective in generating GICs. In this case, the magnetic input was kept constant and the conductivity models under each station were used. The JAT station was chosen to be the reference for the geomagnetic variations. Figure 10 shows the estimated GICs at each station using the rate of change of the magnetic field measured in JAT (identified as B_{JAT}) and the conductivity models of the other stations (Z_{ALF} , Z_{SJC} , and Z_{SMS}). The GICs estimated in JAT (Figure 8) were used again to calculate the relative values associated with the variation in the Earth's conductivity.

The relative values show that GICs are amplified in all stations when compared to JAT. This result is most evident in the high-frequency phase of the March storm, which also indicates that the largest amplifications are observed in ALF. The amplification at all stations in relation to JAT decreases for the other phases of the storms, characterized by events with lower frequency. These results reproduce those shown in Figure 3, where it is observed for the frequencies immediately less than the Nyquist that the highest impedance is observed in ALF and the lowest in JAT. As the frequency decreases, the differences in impedance amplitude also tend to decrease, as also pointed out by the relative values shown in Figure 10.

6. Discussion

The maximum GIC amplitude estimated in this study was about 3.8 A at the ALF equatorial station in central west Brazil during the diurnal third peak of a magnetic storm on 21 and 22 June 2015 ($Dst = -204$ nT).

According to the reliability standards for operation of power transmission networks established by the Institute of Electrical and Electronics Engineers (IEEE, 2015), current oscillations of this order have insignificant influence on transformers. Its major effect could be to convert the transformers into harmonic sources, interfering with other instruments of the electrical networks, especially the protection relays (Gurevich, 2016). However, it should be noted that repeated exposures to even low-level GIC flows can cause cumulative damage that may lead to premature aging of transformers (Gaunt, 2014; Gaunt & Coetzee, 2007; Moodley & Gaunt, 2017). It should be also noted that this storm does not compare with those that occurred in the previous solar cycle and particularly with the great storms recorded in the past that produced the large GIC previously reported in Brazil (Barbosa et al., 2015; Trivedi et al., 1997). In addition, the ALF station is not positioned below the dip equator, where dB_x/dt is still more enhanced. As an example, the third interplanetary pressure pulse of the June storm presented variation rate of 58 nT/min in ALF (magnetic latitude of -2.82°) and ~ 130 nT/min in Huancayo (magnetic latitude of -0.18°), located in western South America. Even considering that Huancayo is affected by coast effects, the difference with the dB_x/dt observed in the vicinity of the dip equator is significant.

Considering the regional significance of these results, in the western region of Brazil, far from possible coastal effects, the dip equator is presently situated over Porto Velho. At this region are located ground electrodes of the ± 600 -KV Rio Madeira high-voltage direct current transmission system. This system crosses 2,375 km in the central region of South America, connecting hydropower plants on the Madeira River (Amazon basin, NW Brazil) to the main consuming areas in SE Brazil. In the Porto Velho region, the transmission line runs approximately parallel to the magnetic meridian and thus it is not expected that large GICs are generated (dB_y/dt is very small in the equatorial region). However, there are secondary lines around Porto Velho running in transversal directions that can be more affected by GIC. Unfortunately, no engineering information is available for these networks. Also, this region is positioned in a mobile belt marginal to cratonic zones, where resistivity is probably not as high as in ALF. Thus, the amplification by the low conductance must be smaller. To the east, the dip equator runs mainly over the Amazon forest, where there are no power lines, and over the large Parnaíba Basin, with high crustal conductance. On this basin, there are transmission lines very close to the dip equator and running approximately parallel to the EEJ. Again, there is no engineering information on these lines.

The GICs amplitudes in stations at low magnetic latitude regions of S-SE Brazil are smaller than those determined for the equatorial region. This is due to both the lower surface impedance under these stations and the higher daytime amplification of the EEJ when compared to SAMA. Nighttime comparison of dB/dt was only possible during the SSC of one of the storms, when larger values were observed in the central part of SAMA. This confirms that the increase in ionospheric conductivity at SAMA occurs both during daytime and nighttime, whereas the EEJ is a typical daytime event. Additional induction effects on dB/dt measured at our stations are negligible due to the 1-D environment and the absence of any huge crustal conductor that can reflect in phase the downward propagating electromagnetic wave.

In spite of the possible dB/dt increase closer to the SAMA center, strong GICs are not expected in this region due to the high electrical conductivity of the crust associated with the depocenters of the Paraná and Chaco-Paraná basins. On the other hand, although less affected by SAMA, JAT, and SJC are located in regions with several power transmission lines. In particular, SJC is positioned in the most populous region of Brazil and, excluding localized sedimentary basins such as the Taubaté basin, this region is covered by metamorphic rocks of high resistivity. Higher surface impedance and possible coast effects can amplify the amplitudes of GICs.

More important for the scope of the present study are the relative values that enable a rough estimate of the individual effects of the Earth's conductivity profiles and the rate of change of the geomagnetic field on the GIC magnitude. As expected, effects related to ground conductance variation are governed by variation of the surface impedance and the sampling rate of the geomagnetic field. For this study, a rate of one sample per minute was used and that only allows us to observe the descending branch of the impedance curve. The effects of the shallower layers appear as a difference in amplitude in these curves, which control the relative effects of the ground conductance between the stations. This difference in amplitude is related to impedance at frequencies above the Nyquist (~ 8.3 mHz in our case), which significantly influences the magnitude of the induced surface electric field (see a more detailed discussion in Grawe et al., 2018).

The effects of the geomagnetic field show that the increase in overhead ionospheric conductivity associated with the EEJ and SAMA amplify the GICs measured at the Earth's surface for geomagnetic variations with frequency higher than 2 mHz. As our analysis took place during the daytime main phases of the magnetic storms, a greater effect of the EEJ was observed. However, the most unanticipated finding was the identification of a cutoff frequency in amplification of the signals by the ionospheric currents. It was observed that signals with frequencies lower than 2 mHz (periods longer than 500 s) are not affected by these currents. This result cannot be associated to any damping effect due to the high conductivity of the chosen reference station (JAT), as it is also observed in the relative values shown in Figure 8. Its interpretation requires a more detailed study of the characteristics of ionospheric currents in EEJ and SAMA during magnetically disturbed periods, which is beyond the scope of this work.

7. Conclusions

A case study is presented on GICs in power grids at sites affected by the EEJ and SAM in Brazil. Using temporal measurements of the ground geomagnetic field and 1-D models of the conductivity structure, GIC magnitudes are estimated in a model Brazilian power network during the two largest magnetic storms recorded so far by the Embrace MagNet network. As the storms are not very severe, the GIC values are small and the main focus of the work is to isolate the effects of the ground conductance and ionospheric conductivity for each site. Spectrograms of the calculated GICs allows quantifying what frequencies are contributing to the signal. The most relevant results are the following:

1. The relative GICs magnitude is controlled by the surface impedance and the sampling rate of the geomagnetic field.
2. Both EEJ and SAMA increase GICs magnitude, with the greatest effects during daytime associated with the EEJ.
3. A cutoff frequency was identified for both EEJ and SAMA; signals with frequencies lower than 2 mHz were not amplified by these regions of enhanced ionospheric conductivity.

References

- Abdu, M. A., Batista, I. S., Carrasco, A. J., & Brum, C. G. M. (2005). South Atlantic Magnetic Anomaly ionization: A review and a new focus on electrodynamic effects in the equatorial ionosphere. *Journal of Atmospheric and Solar-Terrestrial Physics*, 67, 1643–1657. <https://doi.org/10.1016/j.jastp.2005.01.014>
- Astafyeva, E., Zakharenkova, I., & Alken, P. (2016). Prompt penetration electric fields and the extreme topside ionospheric response to the June 22–23, 2015 geomagnetic storm as seen by the Swarm constellation. *Earth, Planets and Space*, 68(1), 152. <https://doi.org/10.1186/s40623-016-0526-x>
- Astafyeva, E., Zakharenkova, I., & Forster, M. (2015). Ionospheric response to the 2015 St. Patrick's Day storm: A global multi-instrumental overview. *Journal of Geophysical Research: Space Physics*, 120, 9023–9037. <https://doi.org/10.1002/2015JA021629>
- Barbosa, C., Alves, L., Caraballo, R., Hartmann, G., Papa, A., & Pirjola, R. (2015). Analysis of geomagnetically induced currents at a low-latitude region over the solar cycles 23 and 24: Comparison between measurements and calculations. *Journal of Space Weather and Space Climate*, 5, A35. <https://doi.org/10.1051/swsc/2015036>
- Bernhardi, E. H., Cilliers, P. J., & Gaunt, C. T. (2008). Improvement in the modelling of geomagnetically induced currents in southern Africa. *South African Journal of Science*, 104, 265–272.
- Bolduc, L. (2002). GIC observations and studies in the Hydro-Quebec power system. *Journal of Atmospheric and Solar-Terrestrial Physics*, 64, 1793–1802. [https://doi.org/10.1016/S1364-6826\(02\)00128-1](https://doi.org/10.1016/S1364-6826(02)00128-1)
- Bologna, M. S., Padilha, A. L., Vitorello, Í., & Pádua, M. B. (2011). Signatures of continental collisions and magmatic activity in central Brazil as indicated by a magnetotelluric profile across distinct tectonic provinces. *Precambrian Research*, 185(1–2), 55–64. <https://doi.org/10.1016/j.precamres.2010.12.003>
- Boteler, D. H. (1994). Geomagnetically induced currents: Present knowledge and future research. *IEEE Transactions on Power Delivery*, 9, 50–58.
- Boteler, D. H., & Pirjola, R. J. (1998). The complex-image method for calculating the magnetic and electric fields produced at the surface of the Earth by the auroral electrojet. *Geophysical Journal International*, 132(1), 31–40. <https://doi.org/10.1046/j.1365-246X.1998.00388.x>
- Brito, P. M. A. (1998). *Sondagens magnetotélúricas transversais à bacia de Taubaté*. Master's thesis Instituto Nacional de Pesquisas Espaciais - INPE. Brazil: São José dos Campos.
- Cagniard, L. (1953). Basic theory of the magneto-telluric method of geophysical prospecting. *Geophysics*, 18, 605–635.
- Caraballo, R., Sánchez Bettucci, L., & Tancredi, G. (2013). Geomagnetically induced currents in the Uruguayan high-voltage power grid. *Geophysical Journal International*, 195, 844–853. <https://doi.org/10.1093/gji/ggt293>
- Carter, B. A., Yizengaw, E., Pradipta, R., Halford, A. J., Norman, R., & Zhang, K. (2015). Interplanetary shocks and the resulting geomagnetically induced currents at the equator. *Geophysical Research Letters*, 42, 6554–6559. <https://doi.org/10.1002/2015GL065060>
- Carter, B., Yizengaw, E., Pradipta, R., Weygand, J., Piersanti, M., Pulkkinen, A., et al. (2016). Geomagnetically induced currents around the world during the 17 March 2015 storm. *Journal of Geophysical Research: Space Physics*, 121, 496–507. <https://doi.org/10.1002/2016JA023344>
- Denardini, C. M., Chen, S. S., Resende, L. C. A., Moro, J., Bilibio, A. V., Fagundes, P. R., et al. (2018). The Embrace Magnetometer Network for South America: Network description and its qualification. *Radio Science*, 53, 288–302. <https://doi.org/10.1002/2017RS006477>

Acknowledgments

The study was supported by fellowships from CNPq (130918/2016-4 and 306390/2017-5). Data for the 2015 geomagnetic storms are available at the Brazilian Studies and Monitoring of Space Weather (EMBRACE/INPE) program (<http://www2.inpe.br/climaespacial/portal/en/>). The DMSP satellites are part of the U.S. Air Force Earth environmental observation program, and the SSUSI UV radiances are provided by the Johns Hopkins University Applied Physics Laboratory (<https://ssusi.jhuapl.edu/>). GIC measurements shown in Figure 4 are available in supporting information of this manuscript. The SYM-H index is provided by the World Data Center for Geomagnetism, Kyoto (<http://wdc.kugi.kyoto-u.ac.jp/>). We thank the national institutes that support the operation of magnetic observatories and INTERMAGNET for promoting high standards of observatory practice. Comments and suggestions by two anonymous reviewers and Editor (Michael Hapgood) helped improve our original manuscript considerably.

- Dragone, G. N. (2018). *A litosfera das Bacias do Chaco-Paraná e Paraná integrando gravimetria e sondagens magnetotéluricas: Novos vínculos à tectônica do Gondwana Sul-Occidental*, Ph.D. thesis Universidade de São Paulo. Brazil: São Paulo.
- Fernandes, C. M., Fontes, S. L., La Terra, E. F., Miquelutti, L. G., & Maurya, V. P. (2018). Electrical crustal structure of Alta Floresta Gold Province eastern sector, SW Amazon Craton, Brazil. *Journal of South American Earth Sciences*, 86, 244–258. <https://doi.org/10.1016/j.jsames.2018.06.022>
- Forbes, J. M. (1981). The equatorial electrojet. *Reviews of Geophysics and Space Physics*, 19, 469–504. <https://doi.org/10.1029/RG019i003p00469>
- Gaunt, C. T. (2014). Reducing uncertainty-responses for electricity utilities to severe solar storms. *Journal of Space Weather and Space Climate*, 4, A01. <https://doi.org/10.1051/swsc/2013058>
- Gaunt, C. T., & Coetzee, G. (2007). Transformer failures in regions incorrectly considered to have low GIC-risk. *2007 IEEE Lausanne Power Tech*, 445, 807–812. <https://doi.org/10.1109/PCT.2007.4538419>
- Grawe, M. A., Makela, J. J., Butala, M. D., & Kamalabadi, F. (2018). The impact of magnetic field temporal sampling on modeled surface electric fields. *Space Weather*, 16, 1721–1739. <https://doi.org/10.1029/2018SW001896>
- Grigoryan, O., Petrov, A., Romashova, V., & Bengin, V. (2005). On the drift of the South Atlantic Anomaly, WDS '05 Proceedings of Contributed Papers, Part II pp. 251–256.
- Gurevich, V. (2016). Solar storms: What is the risk to power transformers? (pp. 25–27). Nooitgedacht, South Africa: EE Publishers (Pty) Ltda.
- IEEE (2015). Guide for establishing power transformer capability while under geomagnetic disturbances, The Institute of Electrical and Electronic Engineering (IEEE). *IEEE Standard C57, 163-2015*, 1–50. <https://doi.org/10.1109/IEEESTD.2015.7286929>
- Kamide, Y., & Kusano, K. (2015). No major solar flares but the largest geomagnetic storm in the present solar cycle. *Space Weather*, 13, 365–367. <https://doi.org/10.1002/2015SW001213>
- Kappenman, J. G. (2003). Storm sudden commencement events and the associated geomagnetically induced current risks to ground-based systems at low-latitude and midlatitude locations. *Space Weather*, 1(3), 1016. <https://doi.org/10.1029/2003SW000009>
- Kikuchi, T., Araki, T., Maeda, H., & Maekawa, K. (1978). Transmission of polar electric fields to the equator. *Nature*, 273, 650–651. <https://doi.org/10.1038/273650a0>
- Lehtinen, M., & Pirjola, R. (1985). Currents produced in earthed conductor networks by geomagnetically-induced electric fields. *Annales Geophysicae*, 3, 479–484.
- Liu, Y. D., Hu, H., Wang, R., Yang, Z., Zhu, B., Liu, Y. A., et al. (2015). Plasma and magnetic field characteristics of solar coronal mass ejections in relation to geomagnetic storm intensity and variability. *The Astrophysical Journal Letters*, 809(2), 1–6. <https://doi.org/10.1088/2041-8205/809/2/L34>
- Liu, C.-M., Liu, L.-G., Pirjola, R., & Wang, Z.-Z. (2009). Calculation of geomagnetically induced currents in mid- to low-latitude power grids based on the plane wave method: A preliminary case study. *Space Weather*, 7, S04005. <https://doi.org/10.1029/2008SW000439>
- Marshall, R. A., Dalzell, M., Waters, C. L., Goldthorpe, P., & Smith, E. A. (2012). Geomagnetically induced currents in the New Zealand power network. *Space Weather*, 10, S08003. <https://doi.org/10.1029/2012SW000806>
- Marshall, R. A., Gorniak, H., van der Walt, T., Waters, C. L., Sciffer, M. D., Miller, M., et al. (2013). Observations of geomagnetically induced currents in the Australian power network. *Space Weather*, 11, 6–16. <https://doi.org/10.1029/2012SW000849>
- Moodley, N., & Gaunt, C. T. (2017). Low energy degradation triangle for power transformer health assessment. *IEEE Transactions on Dielectrics and Electrical Insulation*, 24, 639–646. <https://doi.org/10.1109/TDEI.2016.006042>
- Ngwira, C. M., Pulkkinen, A., McKinnell, L. A., & Cilliers, P. J. (2008). Improved modeling of geomagnetically induced currents in the South African power network. *Space Weather*, 6, S11004. <https://doi.org/10.1029/2008SW000408>
- Ngwira, C. M., Sibeck, D., Silveira, M. V. D., Georgiou, M., Weygand, J. M., Nishimura, Y., & Hampton, D. (2018). A study of intense local dB/dt variations during two geomagnetic storms. *Space Weather*, 16, 676–693. <https://doi.org/10.1029/2018SW001911>
- Padilha, A. L., Alves, M. V., Trivedi, N. B., Kitamura, T.-I., & Shinohara, M. (2003). Bursty P11 activity at the South American equatorial zone during the 29 October 1994 magnetic storm. *Geophysical Research Letters*, 30(19), 2006. <https://doi.org/10.1029/2003GL017999>
- Padilha, A. L., Trivedi, N. B., Vitorello, Í., & da Costa, J. M. (1991). Geophysical constraints on tectonic models of the Taubaté Basin, southeastern Brazil. *Tectonophysics*, 196(1-2), 157–172. [https://doi.org/10.1016/0040-1951\(91\)90294-3](https://doi.org/10.1016/0040-1951(91)90294-3)
- Padilha, A. L., Vitorello, Í., & Rijo, L. (1997). Effects of the equatorial electrojet on magnetotelluric surveys: Field results from northwest Brazil. *Geophysical Research Letters*, 24(1), 89–92. <https://doi.org/10.1029/96GL03792>
- Paulikas, G. A. (1975). Precipitation of particles at low and middle latitudes. *Reviews of Geophysics and Space Physics*, 13, 709–734.
- Paxton, L. J., Morrison, D., Zhang, Y., Kil, H., Wolven, B., Ogorzalek, B. S., et al. (2002). Validation of remote sensing products produced by the Special Sensor Ultraviolet Scanning Imager (SSUSI): A far UV-imaging spectrograph on DMSP F-16. *Proceedings of the SPIE*, 4485, 338–348. <https://doi.org/10.1117/12.454268>
- Pirjola, R. (2000). Geomagnetically induced currents during magnetic storms. *IEEE Transactions on Plasma Science*, 28(6), 1867–1873. <https://doi.org/10.1109/27.902215>
- Pirjola, R. (2002). Review on the calculation of surface electric and magnetic fields and of geomagnetically induced currents in ground-based technological systems. *Surveys in Geophysics*, 23(1), 71–90. <https://doi.org/10.1023/A:1014816009303>
- Pulkkinen, A., Bernabeu, E., Eichner, J., Beggan, C., & Thomson, A. (2012). Generation of 100-year geomagnetically induced current scenarios. *Space Weather*, 10, S04003. <https://doi.org/10.1029/2011SW000750>
- Pulkkinen, A., Thomson, A., Clarke, E., & McKay, A. (2003). April 2000 geomagnetic storm: Ionospheric drivers of large geomagnetically induced currents. *Annales Geophysicae*, 21(3), 709–717. <https://doi.org/10.5194/angeo-21-709-2003>
- Shepherd, S. G. (2014). Altitude-adjusted corrected geomagnetic coordinates: Definition and functional approximations. *Journal of Geophysical Research: Space Physics*, 30, 1–21. <https://doi.org/10.1002/2014JA020264>
- Simpson, F., & Bahr, K. (2005). *Practical magnetotellurics* (1st ed.). United Kingdom: Cambridge University Press, Cambridge.
- Thébault, E., Finlay, C. C., Finlay, C. D., Alken, P., Aubert, J., Aubert, O., et al. (2015). International Geomagnetic Reference Field: The 12th generation. *Earth Planets and Space*, 67, 79. <https://doi.org/10.1186/s40623-015-0228-9>
- Tikhonov, A. N. (1950). On determining the electrical characteristics of the deep layers of the Earth's crust. *Doklady Akademii Nauk SSSR*, 73(2), 295–297.
- Torta, J. M., Serrano, L., Regué, J. R., Sánchez, A. M., & Roldán, E. (2012). Geomagnetically induced currents in a power grid of northeastern Spain. *Space Weather*, 10, S06002. <https://doi.org/10.1029/2012SW000793>
- Trivedi, N. B., Arora, B. R., Padilha, A. L., Costa, J. M., Dutra, S. L. G., Chamalaun, F. H., & Rigoti, A. (1997). Global Pc5 geomagnetic pulsations of March 24, 1991, as observed along the American sector. *Geophysical Research Letters*, 24(13), 1683–1686.
- Trivedi, N. B., Pathan, B. M., Schuch, N. J., Barreto, M., & Dutra, S. L. G. (2005). Geomagnetic phenomena in the South Atlantic anomaly region in Brazil. *Advances in Space Research*, 36(10), 2021–2024. <https://doi.org/10.1016/j.asr.2004.09.020>

- Trivedi, N. B., Vitorello, Í., Kabata, W., Dutra, S. L. G., Padilha, A. L., Bologna, M. S., et al. (2007). Geomagnetically induced currents in an electric power transmission system at low latitudes in Brazil: A case study. *Space Weather*, 5, 04004. <https://doi.org/10.1029/2006SW000282>
- Viljanen, A., Pirjola, R., Prácser, E., Ahmadzai, S., & Singh, V. (2013). Geomagnetically induced currents in Europe: Characteristics based on a local power grid model. *Space Weather*, 11, 575–584. <https://doi.org/10.1002/swe.20098>
- Wait, J. R. (1954). On the relation between telluric currents and the Earth's magnetic field. *Geophysics*, 19(2), 281–289. <https://doi.org/10.1190/1.1437994>
- Watari, S., Kunitake, M., Kitamura, K., Hori, T., Kikuchi, T., Shiokawa, K., et al. (2009). Measurements of geomagnetically induced current in a power grid in Hokkaido, Japan. *Space Weather*, 7, 03002. <https://doi.org/10.1029/2008SW000417>
- Weigel, R. S. (2017). A comparison of methods for estimating the geoelectric field. *Space Weather*, 15, 430–440. <https://doi.org/10.1002/2016SW001504>
- Zhang, J. J., Wang, C., Sun, T. R., Liu, C. M., & Wang, K. R. (2015). GIC due to storm sudden commencement in low-latitude high-voltage power network in China: Observation and simulation. *Space Weather*, 13, 643–655. <https://doi.org/10.1002/2015SW001263>